

4. Find the moment of inertia of a quadrant of the ellipse $(x/a)^2 + (y/b)^2 = 1$ of mass M about the x -axis, if the density at a point is proportional to xy .
5. Find the moment of inertia about the initial line of the cardioid $r = a(1 + \cos \theta)$.
6. Find the moment of inertia of a uniform spherical ball of mass M and radius R about a diameter.
7. Find the moment of inertia of a solid right circular cylinder about (i) its axis (P.T.U., 2006)
(ii) a diameter of the base.
8. Find the M.I. of a solid right circular cone having base-radius r and height h , about (i) its axis, (ii) an axis through the vertex and perpendicular to its axis, (iii) a diameter of its base.
9. Find the moment of inertia of a hollow sphere about a diameter, its external and internal radii being 51 metres and 49 metres.
10. Find the moment of inertia about z -axis of a homogeneous tetrahedron bounded by the planes $x = 0, y = 0, z = x + y$ and $z = 1$.
11. Find the moment of inertia of an octant of the ellipsoid $(x/a)^2 + (y/b)^2 + (z/c)^2 = 1$, about the x -axis.
12. Find the product of inertia of a quadrant of the ellipse $(x/a)^2 + (y/b)^2 = 1$, about the coordinate axes.
13. Show that the principal axes at the origin of the triangle enclosed by $x = 0, y = 0, (x/a) + (y/b) = 1$ are inclined to the x -axis at angles α and $\alpha + \pi/2$, where $\alpha = \frac{1}{2} \tan^{-1} [ab/(a^2 - b^2)]$ (U.P.T.U., 2002)
14. The lengths AB and AD of the sides of a rectangle $ABCD$ are $2a$ and $2b$. Show that the inclination to AB of one of the principal axes at A is $\frac{1}{2} \tan^{-1} \left\{ \frac{3ab}{2(a^2 - b^2)} \right\}$.

7.14 BETA FUNCTION

The beta function is defined as

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx \quad \begin{cases} m > 0 \\ n > 0 \end{cases} \quad \dots(1)$$

Putting $x = 1 - y$ in (1), we get $\beta(m, n) = - \int_1^0 (1-y)^{m-1} y^{n-1} dy$

$$= \int_0^1 y^{n-1} (1-y)^{m-1} dy = \beta(n, m)$$

Thus $\beta(m, n) = \beta(n, m)$... (2)

Putting $x = \sin^2 \theta$ so that $dx = 2 \sin \theta \cos \theta d\theta$, (1) becomes

$$\begin{aligned} \beta(m, n) &= \int_0^{\pi/2} (\sin^2 \theta)^{m-1} (\cos^2 \theta)^{n-1} 2 \sin \theta \cos \theta d\theta \\ &= 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta \end{aligned} \quad \dots(3)$$

which is another form of $\beta(m, n)$.

This function is also *Euler's integral of the first kind**.

7.15 (1) GAMMA FUNCTION

The gamma function is defined as

$$\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx \quad (n > 0) \quad \dots(i)$$

This integral is also known as *Euler's integral of the second kinds*. It defines a function of n for positive values of n .

*After an enormously creative Swiss mathematician *Leonhard Euler (1707-1783)*. He studied under *John Bernoulli* and became a professor of mathematics in St. Petersburg, Russia. Even after becoming totally blind in 1771, he contributed to almost all branches of mathematics.

$$\text{In particular, } \Gamma(1) = \int_0^{\infty} e^{-x} dx = \left[-e^{-x} \right]_0^{\infty} = 1. \quad \dots(ii)$$

(2) Reduction formula for $\Gamma(n)$.

$$\text{Since } \Gamma(n+1) = \int_0^{\infty} e^{-x} x^n dx \text{ [Integrating by parts]} = \left[-x^n e^{-x} \right]_0^{\infty} + n \int_0^{\infty} e^{-x} x^{n-1} dx$$

$$\therefore \Gamma(n+1) = n\Gamma(n) \quad \dots(iii)$$

which is the reduction formula for $\Gamma(n)$. From this formula, it is clear that if $\Gamma(n)$ is known throughout a unit interval say : $1 < n \leq 2$, then the values of $\Gamma(n)$ throughout the next unit interval $2 < n \leq 3$ are found, from which the values of $\Gamma(n)$ for $3 < n \leq 4$ are determined and so on. In this way, the values of $\Gamma(n)$ for all positive values of $n > 1$ may be found by successive application of (iii).

Also using (iii) in the form

$$\Gamma(n) = \frac{\Gamma(n+1)}{n} \quad \dots(iv)$$

We can define $\Gamma(n)$ for values of n for which the definition (1) fails. It gives the value of $\Gamma(n)$ for $0 < n \leq 1$ in terms of the values of $\Gamma(n)$ for $1 < n \leq 2$. Thus we can define $\Gamma(n)$ for all $n < 0$ provided its value for $1 < n \leq 2$ is known. Also if $-1 < n < 0$, (4) gives $\Gamma(n)$ in terms of its values for $0 < n < 1$. Then we may find, $\Gamma(n)$ for $-2 < n < -1$ and so on.

Thus (i) and (iv) together give a complete definition of $\Gamma(n)$ for all values of n except when n is zero or a negative integer and its graph is as shown in Fig. 7.44. The values of $\Gamma(n)$ for $1 < n \leq 2$ are given in (Table I-Appendix 2) from which the values of $\Gamma(n)$ for values of n outside the interval $1 < n \leq 2$ ($n \neq 0, -1, -2, -3, \dots$) may be found.

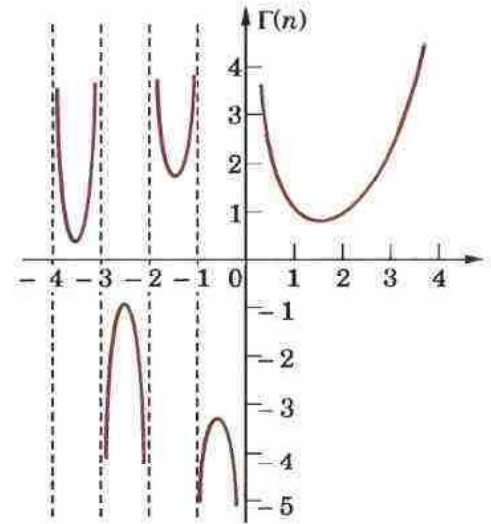


Fig. 7.44

(3) Value of $\Gamma(n)$ in terms of factorial.

Using $\Gamma(n+1) = n\Gamma(n)$ successively, we get

$$\Gamma(2) = 1 \times \Gamma(1) = 1!$$

$$\Gamma(3) = 2 \times \Gamma(2) = 2 \times 1 = 2!$$

$$\Gamma(4) = 3 \times \Gamma(3) = 3 \times 2! = 3!$$

$$\dots\dots\dots$$

In general $\Gamma(n+1) = n!$ provided n is a positive integer

Taking $n = 0$, it defines $0! = \Gamma(1) = 1$.

(4) Value of $\Gamma(\frac{1}{2})$. We have

$$\Gamma\left(\frac{1}{2}\right) = \int_0^{\infty} e^{-x} x^{-1/2} dx$$

$$= 2 \int_0^{\infty} e^{-y^2} dy \text{ which is also } = 2 \int_0^{\infty} e^{-x^2} dx$$

$$\therefore \left[\Gamma\left(\frac{1}{2}\right) \right]^2 = 4 \int_0^{\infty} \int_0^{\infty} e^{-(x^2+y^2)} dx dy = 4 \int_0^{\pi/2} \int_0^{\infty} e^{-r^2} r dr d\theta$$

$$= 4 \cdot \frac{\pi}{2} \int_0^{\infty} e^{-r^2} r dr = 2\pi \left[-\frac{1}{2} e^{-r^2} \right]_0^{\infty} = \pi$$

$$\text{whence } \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} = 1.772 \quad \dots(vi) \quad (\text{V.T.U., 2006})$$

7.16 RELATION BETWEEN BETA AND GAMMA FUNCTIONS

$$\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

$$\text{We have } \Gamma(m) = \int_0^{\infty} e^{-t} t^{m-1} dt \quad [\text{Put } t = x^2 \text{ so that } dt = 2x dx]$$

$$= 2 \int_0^{\infty} e^{-x^2} x^{2m-1} dx \quad \dots(2)$$

Similarly, $\Gamma(n) = 2 \int_0^{\infty} e^{-y^2} y^{2n-1} dy$

$$\begin{aligned} \therefore \Gamma(m)\Gamma(n) &= 4 \int_0^{\infty} e^{-x^2} x^{2m-1} dx \int_0^{\infty} e^{-y^2} y^{2n-1} dy \\ &= 4 \int_0^{\infty} \int_0^{\infty} e^{-(x^2+y^2)} x^{2m-1} y^{2n-1} dx dy \quad \dots(3) \quad [\because \text{the limits of integration are constant.}] \end{aligned}$$

Now change to polar coordinates by writing $x = r \cos \theta$, $y = r \sin \theta$ and $dx dy = r d\theta dr$. To cover the region in (3) which is the entire first quadrant, r varies from 0 to ∞ and θ from 0 to $\pi/2$. Thus (3) becomes

$$\begin{aligned} \Gamma(m)\Gamma(n) &= 4 \int_0^{\pi/2} \int_0^{\infty} e^{-r^2} r^{2(m+n)-1} \cos^{2m-1} \theta \sin^{2n-1} \theta d\theta dr \\ &= \left[2 \int_0^{\pi/2} \cos^{2m-1} \theta \sin^{2n-1} \theta d\theta \right] \times \left[2 \int_0^{\infty} e^{-r^2} r^{2(m+n)-1} dr \right] \quad \dots(4) \end{aligned}$$

But by (2), $2 \int_0^{\infty} e^{-r^2} r^{2(m+n)-1} dr = \Gamma(m+n)$

and by (3) of § 7.14, $2 \int_0^{\pi/2} \cos^{2m-1} \theta \sin^{2n-1} \theta d\theta = \beta(m, n)$.

Thus (4) gives $\Gamma(m)\Gamma(n) = \beta(m, n) \Gamma(m+n)$

(U.T.U., 2010 ; Bhopal, 2009 ; V.T.U., 2008 S)

whence follows (1) which is extremely useful for evaluating definite integrals in terms of gamma functions.

Cor. Rule to evaluate $\int_0^{\pi/2} \sin^p x \cos^q x dx$.

$$\begin{aligned} \int_0^{\pi/2} \sin^p x \cos^q x dx &= \frac{1}{2} \beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right) \quad [\text{By (3) of § 7.14}] \\ &= \frac{\Gamma\left(\frac{p+1}{2}\right) \Gamma\left(\frac{q+1}{2}\right)}{2\Gamma\left(\frac{p+q+2}{2}\right)} \quad \dots(5) \end{aligned}$$

In particular, when $q = 0$, and $p = n$, we have

$$\int_0^{\pi/2} \sin^n x dx = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n+2}{2}\right)} \cdot \frac{\sqrt{\pi}}{2}$$

Similarly,

$$\int_0^{\pi/2} \cos^n x dx = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n+2}{2}\right)} \cdot \frac{\sqrt{\pi}}{2} \quad \dots(6)$$

Example 7.42. Show that

(a) $\Gamma(n) = \int_0^1 \left(\log \frac{1}{y}\right)^{n-1} dy \quad (n > 0).$ (J.N.T.U., 2003 ; Madras, 2003 S)

(b) $\beta(p, q) = \int_0^{\infty} \frac{y^{q-1}}{(1+y)^{p+q}} dy$ (V.T.U., 2003 ; Gauhati, 1999)

$= \int_0^1 \frac{x^{p-1} + x^{q-1}}{(1+x)^{p+q}} dx.$ (V.T.U., 2008 ; Osmania, 2003 ; Rohtak, 2003)

Solution. (a) $\Gamma(n) = \int_0^{\infty} x^{n-1} e^{-x} dx \quad (n > 0)$

$$= \int_1^0 \left(\log \frac{1}{y} \right)^{n-1} y \left(-\frac{1}{y} dy \right) = \int_0^1 \left(\log \frac{1}{y} \right)^{n-1} dy.$$

Put $y = e^{-x}$
i.e., $x = \log (1/y)$
so that $dx = -(1/y) dy$

$$(b) \quad \beta(p, q) = \int_0^1 x^{p-1} (1-x)^{q-1} dx$$

$$= \int_0^1 \frac{1}{(1+y)^{p-1}} \left(\frac{y}{1+y} \right)^{q-1} \frac{-1}{(1+y)^2} dy$$

Put $x = \frac{1}{1+y}$ i.e., $y = \frac{1}{x} - 1$
so that $dx = \frac{-1}{(1+y)^2} dy$

$$= \int_0^\infty \frac{y^{q-1}}{(1+y)^{p+q}} dy = \int_0^1 \frac{y^{q-1}}{(1+y)^{p+q}} dy + \int_1^\infty \frac{y^{q-1}}{(1+y)^{p+q}} dy$$

Now substituting $y = 1/z$ in the second integral, we get

$$\int_1^\infty \frac{y^{q-1}}{(1+y)^{p+q}} dy = \int_1^0 \frac{1}{z^{q-1}} \cdot \frac{1}{(1+1/z)^{p+q}} \left(-\frac{1}{z^2} \right) dz = \int_0^1 \frac{z^{p-1}}{(1+z)^{p+q}} dz.$$

$$\text{Hence, } \beta(p, q) = \int_0^1 \frac{y^{q-1}}{(1+y)^{p+q}} dy + \int_0^1 \frac{z^{p-1}}{(1+z)^{p+q}} dz = \int_0^1 \frac{x^{p-1} + x^{q-1}}{(1+x)^{p+q}} dx.$$

Example 7.43. Express the following integrals in terms of gamma functions :

$$(a) \int_0^1 \frac{dx}{\sqrt{(1-x^4)}}$$

$$(b) \int_0^{\pi/2} \sqrt{(\tan \theta)} d\theta. \quad (\text{Madras, 2006})$$

$$(c) \int_0^\infty \frac{x^c}{e^x} dx \quad (\text{U.P.T.U., 2006})$$

$$(d) \int_0^\infty a^{-bx^2} dx.$$

$$(e) \int_0^1 x^5 [\log (1/x)]^3 dx$$

(Madras, 2000)

Solution. (a) $\int_0^1 \frac{dx}{\sqrt{(1-x^4)}}$

Put $x^2 = \sin \theta$, i.e., $x = \sin^{1/2} \theta$
so that $dx = 1/2 \sin^{-1/2} \theta \cos \theta d\theta$

$$= \int_0^{\pi/2} \frac{1}{2} \cdot \frac{\sin^{-1/2} \theta \cdot \cos \theta d\theta}{\sqrt{(1-\sin^2 \theta)}} = \frac{1}{2} \int_0^{\pi/2} \sin^{-1/2} \theta d\theta = \frac{1}{2} \cdot \frac{\sqrt{\pi}}{2} \cdot \frac{\Gamma\left(-\frac{1}{2}+1\right)}{\Gamma\left(-\frac{1}{2}+2\right)} = \frac{\sqrt{\pi}}{4} \frac{\Gamma\left(\frac{1}{4}\right)}{\Gamma\left(\frac{3}{4}\right)}$$

$$(b) \int_0^{\pi/2} \sqrt{(\tan \theta)} d\theta = \int_0^{\pi/2} \sin^{1/2} \theta \cos^{-1/2} \theta d\theta$$

$$= \frac{\Gamma\left(\frac{1}{2}+1\right) \Gamma\left(-\frac{1}{2}+1\right)}{2\Gamma\left(\frac{1}{2}-\frac{1}{2}+2\right)} = \frac{\Gamma\left(\frac{3}{4}\right) \Gamma\left(\frac{1}{4}\right)}{2\Gamma(1)} = \frac{1}{2} \Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right)$$

$$(c) \int_0^\infty \frac{x^c}{e^x} dx = \int_0^\infty \frac{x^c}{e^{x \log c}} dx$$

$$[\because c^x = e^{\log c^x} = e^{x \log c}]$$

$$= \int_0^\infty e^{-x \log c} x^c dx$$

[Put $x \log c = t$ so that $dx = dt/\log c$]

$$= \int_0^{\infty} e^{-t} \left(\frac{t}{\log c} \right)^c \frac{dt}{\log c} = \frac{1}{(\log c)^{c+1}} \int_0^{\infty} t^c e^{-t} dt = \Gamma(c+1)/(\log c)^{c+1}$$

$$(d) \int_0^{\infty} a^{-bx^2} dx = \int_0^{\infty} e^{-bx^2 \log a} dx \quad \left[\begin{array}{l} \text{Put } (b \log a) x^2 = t \\ \text{so that } dx = dt/2\sqrt{(b \log a)} \end{array} \right]$$

$$= \frac{1}{2\sqrt{(b \log a)}} \int_0^{\infty} e^{-t} t^{-1/2} dt = \frac{\Gamma\left(\frac{1}{2}\right)}{2\sqrt{(b \log a)}} = \frac{\sqrt{\pi}}{2\sqrt{(b \log a)}}$$

$$(e) \int_0^1 x^4 [\log(1/x)]^3 dx = \frac{1}{625} \int_0^{\infty} e^{-t} \cdot t^3 dt \quad \left[\begin{array}{l} \text{Put } x = e^{-t/5} \text{ so that } \log(1/x) = t/5 \\ dx = -\frac{1}{5} e^{-t/5} dt \end{array} \right]$$

$$= \frac{\Gamma(4)}{625} = \frac{6}{625}.$$

Example 7.44. Evaluate $\int_0^{\infty} e^{-ax} x^{m-1} \sin bx dx$ in terms of Gamma function.

(U.P.T.U., 2003)

Solution. We have $\Gamma(m) = \int_0^{\infty} e^{-x} x^{m-1} dx$

[Put $x = ay$, $dx = ady$]

$$= \int_0^{\infty} e^{-ay} a^m y^{m-1} dy \quad \text{or} \quad \int_0^{\infty} e^{-ay} y^{m-1} dy = \Gamma(m)/a^m. \quad \dots(i)$$

Then

$$I = \int_0^{\infty} e^{-ax} x^{m-1} \sin bx dx = \int_0^{\infty} e^{-ax} x^{m-1} (\text{Imaginary part of } e^{ibx}) dx$$

$$= \text{I.P. of } \int_0^{\infty} e^{-(a-ib)x} x^{m-1} dx$$

$$= \text{I.P. of } \{\Gamma(m)/(a-ib)^m\} \quad [\text{By (i)}]$$

$$= \text{I.P. of } \{\Gamma(m)/(r^m (\cos \theta - i \sin \theta)^m)\} \quad \text{where } a = r \cos \theta, b = r \sin \theta$$

$$= \text{I.P. of } \{\Gamma(m)/(r^m (\cos m\theta - i \sin m\theta))\} \quad (\text{Using Demoivre's theorem §19.5})$$

$$= \text{I.P. of } \left\{ \frac{\Gamma(m) \cdot (\cos m\theta + i \sin m\theta)}{r^m (\cos m\theta + i \sin m\theta) (\cos m\theta - i \sin m\theta)} \right\}$$

$$= \frac{\Gamma(m)}{r^m} \sin m\theta \quad \text{where } r = \sqrt{(a^2 + b^2)}, \theta = \tan^{-1} b/a.$$

Example 7.45. Prove that $\int_0^1 \frac{x^2 dx}{\sqrt{(1-x^4)}} \times \int_0^1 \frac{dx}{\sqrt{(1+x^4)}} = \frac{\pi}{4\sqrt{2}}$

$$\text{Solution. } \int_0^1 \frac{x^2 dx}{\sqrt{(1-x^4)}} \times \int_0^{\pi/2} \frac{\sin \theta}{\cos \theta} \cdot \frac{\cos \theta}{2\sqrt{(\sin \theta)}} d\theta \quad \left[\text{Putting } x^2 = \sin \theta, dx = \frac{\cos \theta d\theta}{2\sqrt{(\sin \theta)}} \right]$$

$$= \frac{1}{2} \int_0^{\pi/2} \sqrt{(\sin \theta)} d\theta = \frac{1}{4} \beta\left(\frac{3}{4}, \frac{1}{2}\right) = \frac{1}{4} \frac{\Gamma(3/4) \Gamma(1/2)}{\Gamma(5/4)} = \frac{\Gamma(3/4) \Gamma(1/2)}{\Gamma(1/4)}$$

$$\int_0^1 \frac{dx}{\sqrt{(1+x^4)}} = \int_0^{\pi/4} \frac{\sec^2 \theta d\theta}{2\sqrt{(\tan \theta) \sec \theta}} \quad \left[\text{Putting } x^2 = \tan \theta, dx = \frac{\sec^2 \theta d\theta}{2\sqrt{(\tan \theta)}} \right]$$

$$= \frac{1}{\sqrt{2}} \int_0^{\pi/4} \frac{d\theta}{\sqrt{(\sin 2\theta)}} = \frac{1}{2\sqrt{2}} \int_0^{\pi/2} \sin^{-1/2} \phi d\phi \quad \left[\text{Putting } 2\theta = \phi, d\theta = \frac{1}{2} d\phi \right]$$

$$= \frac{1}{4\sqrt{2}} \beta\left(\frac{1}{4}, \frac{1}{2}\right) = \frac{1}{4\sqrt{2}} \frac{\Gamma(1/4) \Gamma(1/2)}{\Gamma(3/4)}$$

$$\therefore \int_0^1 \frac{x^2 dx}{\sqrt{1-x^4}} \times \int_0^1 \frac{dx}{\sqrt{1+x^4}} = \frac{1}{4\sqrt{2}} \left\{ \Gamma\left(\frac{1}{2}\right) \right\}^2 = \frac{\pi}{4\sqrt{2}}.$$

Example 7.46. Prove that (i) $\beta(m, 1/2) = 2^{2m-1} \beta(m, m)$ (V.T.U., 2004)

$$(ii) \Gamma(m) \Gamma\left(m + \frac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{2m-1}} \Gamma(2m) \quad (\text{Duplication Formula})$$

(V.T.U., 2010 ; Kerala, M.E., 2005 ; Madras, 2003 S)

Solution. (i) We know that $\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$... (1)

Putting $n = \frac{1}{2}$, we have $\beta\left(m, \frac{1}{2}\right) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta d\theta$... (2)

Again putting $n = m$ in (i), we get $\beta(m, m) = 2 \int_0^{\pi/2} (\sin \theta \cos \theta)^{2m-1} d\theta$

$$\begin{aligned} &= \frac{1}{2^{2m-2}} \int_0^{\pi/2} \sin^{2m-1} 2\theta d\theta \\ &= \frac{1}{2^{2m-1}} \int_0^{\pi} \sin^{2m-1} \phi d\phi, \text{ putting } 2\theta = \phi \\ &= \frac{1}{2^{2m-1}} \cdot 2 \int_0^{\pi/2} \sin^{2m-1} \phi d\phi \end{aligned}$$

or $2^{2m-1} \beta(m, m) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta d\theta = \beta\left(m, \frac{1}{2}\right)$ [by (2)]

(ii) Rewriting the above result in terms of Γ functions, we get

$$2^{2m-1} \frac{\Gamma(m) \Gamma(m)}{\Gamma(m+m)} = \frac{\Gamma(m) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(m + \frac{1}{2}\right)} \quad \left[\because \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \right]$$

or $\Gamma(m) \Gamma\left(m + \frac{1}{2}\right) = \frac{\sqrt{\pi} \Gamma(2m)}{2^{2m-1}}.$

Example 7.47. Prove that

$$(a) \iint_D x^{l-1} y^{m-1} dx dy = \frac{\Gamma(l) \Gamma(m)}{\Gamma(l+m+1)} h^{l+m} \text{ where } D \text{ is the domain } x \geq 0, y \geq 0 \text{ and } x+y \leq h.$$

(U.P.T.U., 2005)

$$(b) \iiint_V x^{l-1} y^{m-1} z^{n-1} dx dy dz = \frac{\Gamma(l) \Gamma(m) \Gamma(n)}{\Gamma(l+m+n+1)}$$

where V is the region $x \geq 0, y \geq 0, z \geq 0$ and $x+y+z \leq 1$. This important result is known as **Dirichlet's integral***.

Solution. (a) Putting $x/h = X$ and $y/h = Y$, we see that the given integral

$$\begin{aligned} &= \iint_{D'} (hX)^{l-1} (hY)^{m-1} h^2 dXdY \text{ where } D' \text{ is the domain } X \geq 0, Y \geq 0 \text{ and } X+Y \leq 1. \\ &= h^{l+m} \int_0^1 \int_0^{1-X} X^{l-1} Y^{m-1} dY dX = h^{l+m} \int_0^1 X^{l-1} \left[\frac{Y^m}{m} \right]_0^{1-X} dX \\ &= \frac{h^{l+m}}{m} \int_0^1 X^{l-1} (1-X)^m dX = \frac{h^{l+m}}{m} \beta(l, m+1) = \frac{h^{l+m}}{m} \cdot \frac{\Gamma(l) \Gamma(m+1)}{\Gamma(l+m+1)} \end{aligned}$$

*Named after a German mathematician *Peter Gustav Lejeune Dirichlet* (1805–1859) who studied under Cauchy and succeeded Gauss at Gottingen. He is known for his contributions to Fourier series and number theory.

$$= h^{l+m} \frac{\Gamma(l) \Gamma(m)}{\Gamma(l+m+1)} \quad \dots(i) [\because \Gamma(m+1)/m = \Gamma(m)]$$

(b) Taking $y + z \leq 1 - x (= h)$: say, the triple integral

$$\begin{aligned} &= \int_0^1 \int_0^{1-x} \int_0^{1-x-y} x^{l-1} y^{m-1} z^{n-1} dz dy dx \\ &= \int_0^1 x^{l-1} \left[\int_0^h \int_0^{h-y} y^{m-1} z^{n-1} dz dy \right] dx = \int_0^1 x^{l-1} \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n+1)} h^{m+n} dx \quad \dots [\text{By (i)}] \\ &= \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n+1)} \int_0^1 x^{l-1} (1-x)^{m+n} dx = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n+1)} \beta(l, m+n+1) \\ &= \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n+1)} \cdot \frac{\Gamma(l) \Gamma(m+n+1)}{\Gamma(l+m+n+1)} = \frac{\Gamma(l) \Gamma(m) \Gamma(n)}{\Gamma(l+m+n+1)}. \end{aligned}$$

Example 7.48. Evaluate the integral $\iiint x^{l-1} y^{m-1} z^{n-1} dx dy dz$ where x, y, z are all positive with condition, $(x/a)^p + (y/b)^q + (z/c)^r \leq 1$. (U.P.T.U., 2005 S)

Solution. Put $(x/a)^p = u$, i.e., $x = au^{1/p}$ so that $dx = \frac{a}{p} u^{1/p-1} du$

$(y/b)^q = v$, i.e., $y = bv^{1/q}$ so that $dy = \frac{b}{q} v^{1/q-1} dv$

and $(z/c)^r = w$, i.e., $z = cw^{1/r}$ so that $dz = \frac{c}{r} w^{1/r-1} dw$

Then
$$\begin{aligned} &\iiint x^{l-1} y^{m-1} z^{n-1} dx dy dz \\ &= \iiint (au^{1/p})^{l-1} (bv^{1/q})^{m-1} (cw^{1/r})^{n-1} \left(\frac{a}{p}\right) u^{1/p-1} \left(\frac{b}{q}\right) v^{1/q-1} \left(\frac{c}{r}\right) w^{1/r-1} du dv dw \\ &= \frac{a^l b^m c^n}{pqr} \iiint u^{l/p-1} v^{m/q-1} w^{n/r-1} du dv dw \text{ where } u + v + w \leq 1. \\ &= \frac{a^l b^m c^n}{pqr} \frac{\Gamma(l/p) \Gamma(m/q) \Gamma(n/r)}{\Gamma(l/p + m/q + n/r + 1)} \quad [\text{By Dirichlet's integral}] \end{aligned}$$

Example 7.49. The plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ meets the axes in A, B and C . Apply Dirichlet's integral to find the volume of the tetrahedron $OABC$. Also find its mass if the density at any point is $kxyz$. (U.P.T.U., 2004)

Solution. Put $x/a = u, y/b = v, z/c = w$ then the tetrahedron $OABC$ has $u \geq 0, v \geq 0, w \geq 0$ and $u + v + w \leq 1$.

$$\begin{aligned} \therefore \text{ volume of this tetrahedron} &= \iiint_D dx dy dz \\ &= \iiint_D abc du dv dw \quad \left[\begin{array}{l} a dx = a du, dy = b dv, dz = c dw \\ \text{for } D' = u \geq 0, v \geq 0, w \geq 0 \text{ \& } u + v + w \leq 1. \end{array} \right. \\ &= abc \iiint_D u^{1-1} v^{1-1} w^{1-1} du dv dw \\ &= abc \frac{\Gamma(1) \Gamma(1) \Gamma(1)}{\Gamma(1+1+1+1)} = \frac{abc}{6} \quad [\text{By Dirichlet's integral}] \\ \text{Mass} &= \iiint kxyz dx dy dz = \iiint k(au)(bv)(cw) abc du dv dw \\ &= ka^2 b^2 c^2 \iiint u^{2-1} v^{2-1} w^{2-1} du dv dw \\ &= ka^2 b^2 c^2 \frac{\Gamma(2) \Gamma(2) \Gamma(2)}{\Gamma(2+2+2+1)} ka^2 b^2 c^2 \cdot \frac{1}{6!} = \frac{k}{720} a^2 b^2 c^2. \end{aligned}$$

PROBLEMS 7.8

1. Compute :

(i) $\Gamma(3.5)$ (Assam, 1998)

(ii) $\Gamma(4.5)$

(iii) $\Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right)$ (S.V.T.U., 2009) (iv) $\beta(2.5, 1.5)$ (v) $\beta\left(\frac{9}{2}, \frac{7}{2}\right)$. (Andhra, 2000)

2. Express the following integrals in terms of gamma functions :

(i) $\int_0^\infty e^{-x^2} dx$ (ii) $\int_0^\infty x^{p-1} e^{-kx} dx (k > 0)$ (Delhi, 2002 ; V.T.U., 2000)

(iii) $\int_0^\infty \sqrt{x} e^{-x^3} dx$ (J.N.T.U., 2003) (iv) $\int_0^\infty \frac{dx}{x^{p+1} (x-1)^q} (-p < q < 1)$

3. Show that :

(i) $\int_0^\infty \frac{x^4}{4^x} dx = \frac{\Gamma(5)}{(\log 4)^5}$ (Marathwada, 2008)

(ii) $\int_0^{\pi/2} \sqrt{\cot \theta} d\theta = \frac{1}{2} \Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right)$ (Osmania, 2003 S ; V.T.U., 2001)

(iii) $\int_0^{\pi/2} [\sqrt{\tan \theta} + \sqrt{\sec \theta}] d\theta = \frac{1}{2} \Gamma\left(\frac{1}{4}\right) \left\{ \Gamma\left(\frac{3}{4}\right) + \sqrt{\pi} \Gamma\left(\frac{3}{4}\right) \right\}$ (J.N.T.U., 2000)

(iv) $\int_0^{\pi/2} \sqrt{\sin \theta} d\theta \times \int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} = \pi$. (V.T.U., 2007)

4. Given $\int_0^\infty \frac{x^{n-1}}{1+x} dx = \frac{\pi}{\sin n\pi}$, show that $\Gamma(n) \Gamma(1-n) = \frac{\pi}{\sin n\pi}$. (S.V.T.U., 2008)

Hence evaluate $\int_0^\infty \frac{dy}{1+y^4}$. (V.T.U., 2006 ; J.N.T.U., 2005)

5. Prove that :

(i) $\int_0^1 \frac{x dx}{\sqrt{1-x^5}} = \frac{1}{5} \beta\left(\frac{2}{5}, \frac{1}{2}\right)$ (Raipur, 2006) (ii) $\int_0^1 \frac{dx}{\sqrt{1+x^4}} = \frac{1}{4\sqrt{2}} \beta\left(\frac{1}{4}, \frac{1}{2}\right)$. (V.T.U., 2003)

(iii) $\int_0^1 x^3 (1-\sqrt{x})^5 dx = 2\beta(8, 6)$. (Marathwada, 2008 ; J.N.T.U., 2006)

6. Show that (i) $\int_a^b (x-a)^{m-1} (b-x)^{n-1} dx = (b-a)^{m+n-1} \beta(m, n)$ (P.T.U., 2010 ; Mumbai, 2005)

(ii) $\int_0^\infty \frac{x^{m-1}}{(a+bx)^{m+n}} dx = \frac{1}{a^n b^m} \beta(m, n)$ (Nagpur, 2009) (iii) $\int_0^\infty \frac{x^{10} - x^{18}}{(1+x)^{30}} dx = 0$ (Mumbai, 2005)

(iv) $\int_0^1 \frac{(1-x^4)^{3/4}}{(1+x^4)^2} dx = \frac{1}{2^{9/2}} \beta\left(\frac{7}{4}, \frac{1}{4}\right)$ (Mumbai, 2007)

7. Prove that $\int_0^1 x^m (\log x)^n dx = \frac{(-1)^n n!}{(m+1)^{n+1}}$, where n is a positive integer and $m > -1$. (S.V.T.U., 2006)

Hence evaluate $\int_0^1 x (\log x)^3 dx$. (Nagpur, 2009)

8. Show that $\int_0^1 y^{q-1} \left(\log \frac{1}{y}\right)^{p-1} dy = \frac{\Gamma(p)}{q^p}$, where $p > 0, q > 0$. (Rohtak, 2006 S)

9. Express $\int_0^1 x^m (1-x^n)^p dx$ in terms of gamma functions (Marathwada, 2008)

Hence evaluate : (i) $\int_0^1 x(1-x^3)^{10} dx$. (Bhopal, 2008) (ii) $\int_0^1 \frac{dx}{\sqrt{1-x^n}}$ (Anna, 2005)

10. Prove that $\int_0^\infty \frac{dx}{(e^x + e^{-x})^n} = \frac{1}{4} \beta\left(\frac{n}{2}, \frac{n}{2}\right)$ and hence evaluate $\int_0^\infty \operatorname{sech}^8 x \, dx$.

11. Prove that $\beta\left(n + \frac{1}{2}, n + \frac{1}{2}\right) = \frac{\Gamma(n + 1/2)\sqrt{\pi}}{2^{2n} \Gamma(n+1)}$. Hence show that $2^n \Gamma(n + 1/2) = 1 \cdot 3 \cdot 5 \dots (2n-1) \sqrt{\pi}$

(Mumbai, 2007)

12. Prove that :

$$(i) \frac{\beta(m+1, n)}{m} = \frac{\beta(m, n+1)}{n} = \frac{\beta(m, n)}{m+n}$$

$$(ii) \beta(n, n) = \frac{\sqrt{\pi} \Gamma(n)}{2^{2n-1} \Gamma\left(n + \frac{1}{2}\right)}$$

$$(iii) \Gamma\left(n + \frac{1}{2}\right) = \frac{\Gamma(2n+1) \sqrt{\pi}}{2^{2n} \Gamma(n+1)}$$

$$(iv) \beta(m+1) + \beta(m, n+1) = \beta(m, n)$$

(Bhopal, 2008 ; J.N.T.U., 2006 ; Madras, 2003)

13. Show that $\iint x^{m-1} y^{n-1} dx \, dy$ over the positive quadrant of the ellipse

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 1 \text{ is } \frac{a^m b^n}{2n} \beta\left(\frac{m}{2}, \frac{n}{2} + 1\right).$$

14. Show that the area in the first quadrant enclosed by the curve $(x/a)^\alpha + (y/b)^\beta = 1$, $\alpha > 0$, $\beta > 0$, is given by

$$\frac{ab}{\alpha + \beta} \frac{\Gamma(1/\alpha) \Gamma(1/\beta)}{\Gamma(1/\alpha + 1/\beta)}.$$

15. Find the mass of an octant of the ellipsoid $(x/a)^2 + (y/b)^2 + (z/c)^2 = 1$, the density at any point being $\rho = kxyz$.

(U.P.T.U., 2002)

7.17 (1) ELLIPTIC INTEGRALS

In Applied Mathematics, we often come across integrals of the form $\int_0^1 e^{-x^2} dx$ or $\int_0^1 \sin x^2 dx$ which cannot be evaluated by any of the standard methods of integration. In such cases, we may find the value to any desired degree of accuracy by expanding their integrands as power series. An important class of such integrals is the *elliptic integrals*.

Def. The integral $F(k, \phi) = \int_0^\phi \frac{dx}{\sqrt{1 - k^2 \sin^2 x}} \quad (k^2 < 1)$... (i)

which is a function of the two variables k and ϕ , is called the *elliptic integral of the first kind with modulus k and amplitude ϕ* .

The integral $E(k, \phi) = \int_0^\phi \sqrt{1 - k^2 \sin^2 x} \, dx \quad (k^2 < 1)$... (ii)

is called the *elliptic integral of the second kind with modulus k and amplitude ϕ* .

The name *elliptic integral* arose from its original application in finding the length of an elliptic arc (Fig. 7.45). For instance, consider the ellipse

$$x = a \cos \phi, \quad y = b \sin \phi, \quad (a < b)$$

Then length of its arc

$$\begin{aligned} AP &= \int_0^\phi \sqrt{\left(\frac{dx}{d\phi}\right)^2 + \left(\frac{dy}{d\phi}\right)^2} d\phi = \int_0^\phi \sqrt{[(-a \sin \phi)^2 + (b \cos \phi)^2]} d\phi \\ &= \int_0^\phi \sqrt{[b^2 + (a^2 - b^2) \sin^2 \phi]} d\phi = b \int_0^\phi \sqrt{1 - \left(1 - \frac{a^2}{b^2}\right) \sin^2 \phi} d\phi \\ &= bE(k, \phi) \text{ for } k^2 = 1 - a^2/b^2 \leq 1. \end{aligned}$$

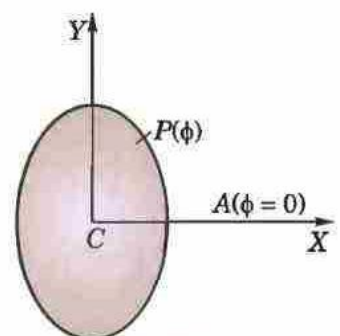


Fig. 7.45